Water-budget lab • evidence

Soil: account for the water

Name ______________________________ Date ________________

Constructed sample data. Same 100 cm³ loosely filled starting soil volume, matched moisture, pots/filters. Add 60 mL over 10 seconds. Collect at 5 minutes from pour start. Assume no spills and negligible evaporation and negligible initial water.

Soil	Input (mL)	Collected at 5 min (mL)
A	60	46
B	60	34
C	60	22

P • Clean run

Same apparatus and method. No spills observed; initial water and evaporation are ignored.

Q • Standing water

A puddle of water remains above the soil at 5 minutes.

R • Spill

Some added water escapes onto the tray and is not collected.

W1. For B, choose 60−34, 60+34 or 34−60. Calculate the amount not collected.

W2. Why is “exactly 38 mL held inside soil C” too certain?

Route 1 • Follow the water

Use the constructed sample data. A calculator or an adult scribe is welcome.

S1. A receives 60 mL and collects 46 mL. Which calculation finds what was not collected?

A. 60−46

B. 60+46

S2. Calculate or use a calculator:60−46=__mL.

The calculated amounts not collected are A 14 mL, B 26 mL and C 38 mL. These come from the constructed sample data; they are setup estimates at 5 minutes.

S3. Which sample has the greatest amount not collected? Use one number.

S4. Name or draw somewhere outside the soil that water could remain.

S5. Complete: This does / does not tell us exactly how much water roots can use, because __.

Route 2 • Record and compare

Use the constructed data or actual readings from an approved practical. Write the data source. Optional spreadsheet: put Soil in A1, Input_mL in B1, Collected_mL in C1, Not_collected_mL in D1. In D2 enter=B2-C2 and copy down.

T1. Data source: __. Our fixed time: __. One other matched condition: __.

T2: Soil	Input(mL)	Collected(mL)	Not collected(mL)
A			
B			
C			

T3. Which sample has the greatest remainder? Compare at least two numbers.

T4. Write a careful conclusion using “at 5 minutes” and “not collected”.

T5. Describe one thing to check before calling the remainder water held by the soil.

Route 3 • Test the water account

Use the constructed five-minute dataset on the evidence page.

X1. Calculate all three amounts not collected and rank them.

Constructed spill case:60 mL added; 30 mL collected; 10 mL spilled onto a separate tray. Assume no evaporation or initial water.

X2. How much was not collected? How much could remain in the setup after allowing for the spill?

Constructed longer test: sample C collects 22 mL by 5 minutes and 30 mL by 10 minutes in total. Input stays 60 mL; no spills; initial water and evaporation ignored.

X3. Find the amount not collected at each time. What changed?

X4. Why cannot this simple test tell us exactly how much water plant roots can use?

Exit and reflection

Use words, numbers, a diagram or a spoken response. A calculator is allowed.

E1. 60 mL goes in; 40 mL is collected. How much was not collected?

E2. Name a place outside the soil where some water may remain.

E3. Why should a result include the collection time?

R1. A careful phrase I added to my conclusion was __.

R2. A next question about soil and water is __.